

Problem Instructions

{ Question Description }

Approach the problem methodically. Ensure all conclusions are based on precise calculations and logical deductions. Feel free to explore various solution methods and cross-check results for consistency. Maintain dynamic thinking and always verify each step of your reasoning.

Present the final answer in `\boxed{ }` format, like this: $\boxed{\text{ANSWER}}$, where ANSWER is the final result or expression.

Think carefully and break down the problem step by step.